



Part—9

LISP TEARS OF JOY

Lisp has been hailed as the world's most powerful programming language. But only the top percentile of programmers use it, because of its cryptic syntax and academic reputation. This is rather unfortunate, since Lisp isn't that hard to grasp. If you want to be among the *crème de la crème*, this series is for you. This is the ninth article in the series that began in the June 2011 issue of *LFY*.

A popular myth about Common Lisp is that it does not support a GUI. This is not true. Lisp systems have supported the GUI since the late 1970s—much before low-cost consumer computers adopted it. The Xerox Alto, developed at Xerox PARC in 1973, was the first computer to use the desktop metaphor and a mouse-driven GUI. It was not a commercial product. Later, Xerox Star was introduced by Xerox Corporation in 1981. This was the first commercial system to incorporate the GUI, and it came with Lisp and SmallTalk for the research and software development market (the Apple Macintosh, released in 1984, was the first commercially *successful* product to use a multi-panel window GUI).

Windows and Macintosh users occasionally find the Lisp GUI coarse and a bit unfamiliar. Several commercial Lisp environments offer graphical interface builders that let you build widgets with point/click and drag/drop techniques, but the look and feel is often 'plain'. These Lisp environments typically provide wrappers around the collection of graphic routines supported by the OS, such that these wrappers can be used from within your Lisp program.

Examples using LispWorks and CAPI tool-kit

LispWorks (available from <http://www.LispWorks.com/>) is an IDE for ANSI Common Lisp. It runs on Linux, Mac OS X, Windows and other operating systems. It uses Common Application Programmer's Interface (CAPI),